

Empathy Map Template

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	4 Marks

Empathy Map Framework:

An empathy map provides deep insight into user behaviors, motivations, and pain points when engaging with AI FitTrack API for activity logging and AI fitness intelligence.

Quadrant	User Quotes & Cognitive Observations	Technical Alignment in AI FitTrack API
SAYS (Verbal Quotes)	• 'I need a fast API to log my daily cardio and strength routines.' • 'Can AI generate a workout plan suited for my age and experience level?' • 'I want clean JSON responses without bloated client subscriptions.'	Node.js Express REST API with structured JSON bodies, JWT Bearer authentication, and Google Gemini AI recommendation endpoints.
THINKS (Beliefs & Doubts)	• 'Will the AI workout recommendations really fit my fitness goal?' • 'Are my workout logs strictly isolated and private from other users?' • 'Is the response time fast enough for mobile applications?'	Customized prompt constraints in geminiService.js; User-scoped ObjectId queries in MongoDB; Low-latency responses under 100ms for CRUD and 1.4s for AI.
DOES (Observed Actions)	• Registers user profile and secures JWT auth token. • Logs workouts specifying category, duration (mins), and calories burned. • Queries workouts by date or keyword search (?q=cardio). • Requests AI workout plans and fitness insights via POST endpoints.	Endpoints POST /api/auth/register, POST /api/workouts, GET /api/workouts/search, POST /api/ai/workout-recommendation, and POST /api/ai/fitness-insights.
FEELS (Emotional State)	• Motivated when receiving instant personalized fitness routines. • Reassured by encrypted passwords and protected endpoints. • Relieved having all workout history queryable in a unified backend.	High developer confidence with production-ready error handling, 100% test pass rate in Postman, and robust data persistence in MongoDB Atlas.

Pains & Gains Analysis:

Dimension	Identified User Friction & Aspirations	AI FitTrack API Architectural Solution
PAINS (Friction Points)	Rigid subscription paywalls, generic static workouts, manual calorie math, complex LLM prompting setups.	Completely open REST architecture, turnkey Google Gemini AI integration, automated input validation and calorie tracking.
GAINS (Value Realized)	Instant tailored workout routines, automated progress insights, secure JWT data isolation, clean developer docs.	Tailored workout generation in 1.4s, comprehensive Postman test suite, robust Mongoose schemas, and full developer README.